EXPERIMENT 12

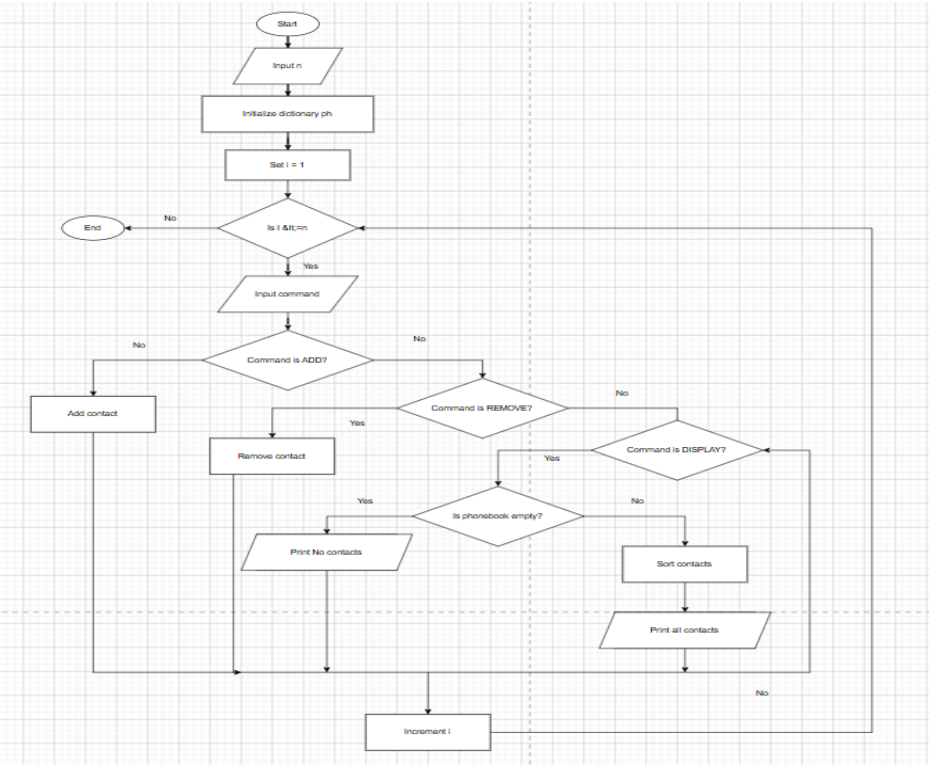
12.1.1//Phone num e database management

Ans: ALGORITHM:

Step 1: Start
Step 2: Read integer n (number of commands)
Step 3: Create an empty dictionary ph
Step 4: Repeat steps for i = 1 to n

- Read command input and split into list cmd
- If cmd[0] == "ADD"
    - Add contact → ph[name] = number
- Else if cmd[0] == "REMOVE"
    - Remove contact using pop()
- Else if cmd[0] == "DISPLAY"
    - If dictionary is empty → Print **"No contacts"**
    - Else → Sort dictionary keys and print **name:number**
      Step 5: Stop

FLOWCHART:



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12.1.1. Phone Number Database Management

00:47

Write a Python program to manage a phone number database using a dictionary. The dictionary stores contact names as keys and phone numbers as values.

The program should support three operations: adding a contact (if the name already exists, update the phone number), removing a contact (if the name does not exist, ignore the operation), and displaying all contacts in alphabetical order by name.

Input Format:

1. First line contains an integer n representing the number of operations.
2. Each of the next n lines contains a single operation in one of the following formats:
   - "ADD <name> <phone_number>" where <name> is a string, <phone_number> is a string of 10 digits.
   - "REMOVE <name>" - name is a string.
   - "DISPLAY" - no additional parameters.

Output Format:

- For display operation, print all contacts in alphabetical order by name in the format:

```
<name>: <phone_number>
```

- If the database is empty during display operation, print: "No contacts".
- Do not print anything for add and remove operations.

Constraints:

- Name contains only alphabetic characters (no spaces). And it is case sensitive.

Explorer phoneDat... Submit Debugger

```python
n = int(input())
ph={}

for i in range(n):
    CMD = input().split()

    if CMD[0] == "ADD":
        ph[CMD[1]] = CMD[2]

    elif CMD[0] == "REMOVE":
        if CMD[1] in ph:
```

Average time 0.015 s 15.25 ms | Maximum time 0.021 s 21.00 ms

5 out of 5 shown test case(s) passed

3 out of 3 hidden test case(s) passed

Test case 1 18 ms Debug

Expected output	Actual output
4	4
ADD Alice 9876543210	ADD Alice 9876543210
ADD Bob 8765432109	ADD Bob 8765432109
ADD Charlie 7777777777	ADD Charlie 7777777777
DISPLAY	DISPLAY
Alice: 9876543210	Alice: 9876543210

Terminal Test cases

Prev Reset Submit Next